CocoConnect

Flutter Project Configuration Guide

For team members · Group 38 · Prepared by: S.N. Vidanage (2023/IS/108)

1. Purpose

This guide gives every team member the same Flutter setup, so the project builds and runs identically on each machine. The dependency files referred to here are included alongside this guide in the Dependency Files folder.

2. Required Tools

Tool	Version	Check with
Flutter SDK	3.44 or newer (stable channel)	flutter --version
Dart SDK	3.12 or newer (comes with Flutter)	dart --version
Android Studio + Android SDK	Recent version, with platform tools	flutter doctor
Java (for Android builds)	JDK 17	java -version
Android emulator	Pixel 7, API 35 (or a physical Android phone)	flutter devices
Node.js + npm (for the backend)	Node.js 18 or newer	node -v
Git	Any recent version	git --version

Note: an older Flutter fails with "The current Dart SDK version is … because kapconnect_mobile requires SDK version ^3.12.2". Run flutter upgrade.

3. Project Settings (pubspec.yaml)

Setting	Value	Meaning
name	kapconnect_mobile	Package name used in imports
version	1.0.0+1	App version and build number
publish_to	none	Private project, never published to pub.dev
environment.sdk	^3.12.2	Minimum Dart version
uses-material-design	true	Material icons and widgets
assets	assets/services/	Images bundled with the app

4. Dependencies

Package	Version	Used for
http	^1.6.0	Calling the backend REST API
shared_preferences	^2.5.6	Keeping the login token on the phone
flutter_map	^8.3.2	OpenStreetMap maps
latlong2	^0.10.1	Map coordinates
geolocator_android	^5.1.1+1	Phone GPS on Android
geolocator_platform_interface	^4.4.0	GPS interface used by the location service
cupertino_icons	^1.0.8	iOS-style icons
flutter_lints (dev)	^6.0.0	Recommended code quality rules

- pubspec.lock fixes the exact versions, so everyone gets the same packages. Commit it with any dependency change.
- Only the Android geolocator package is used, not the full geolocator plugin, because the full plugin needs Windows Developer Mode (symbolic links) to build on Windows.

5. Android Configuration

File	Setting	Why
AndroidManifest.xml	INTERNET permission	Talk to the backend
AndroidManifest.xml	ACCESS_FINE_LOCATION, ACCESS_COARSE_LOCATION	GPS location at sign-up and on the map
AndroidManifest.xml	usesCleartextTraffic="true"	Allow plain HTTP to the local development backend
app/build.gradle.kts	Java and Kotlin target 17	Required by the current Android Gradle plugin
app/build.gradle.kts	minSdk / targetSdk from Flutter	Kept in step with the Flutter version
gradle.properties	kotlin.incremental=false	Fixes Kotlin build errors on Windows when the pub cache (C:) and the project (another drive) are on different drives

6. Code Quality

- analysis_options.yaml includes package:flutter_lints/flutter.yaml.
- Run flutter analyze before every merge; fix new warnings in your own files.

7. Connecting to the Backend

Where the app runs	Backend address
Android emulator	http://10.0.2.2:5000 (the emulator's name for the computer's localhost) – default
Web (Chrome)	http://localhost:5000 – default
Physical phone on the same Wi-Fi	flutter run --dart-define=API_URL=http://<computer-IP>:5000

8. Project Structure

```
lib/main.dart        entry point
lib/screens/         screens (buyer/, orders/, chat/ ...)
lib/services/        API calls (api_client.dart and one service per area)
lib/widgets/         reusable components
assets/services/     bundled images
backend/             Node.js + Express API
```

9. First-Time Setup

```
git clone <repository> && cd CocoConnect
git checkout integration-senuji
flutter pub get
flutter doctor              # fix anything marked X
cd backend && npm install   # create backend/.env from .env.example
```

10. Daily Workflow

```
git pull
cd backend && npm run dev   # terminal 1
flutter run                 # terminal 2, emulator running
                            # r = hot reload, R = hot restart, q = quit
```

11. Common Problems

Problem	Fix
'adb' is not recognised	Add <Android SDK>\platform-tools to PATH, or run adb with its full path
Emulator shows as unauthorized	adb kill-server, then adb start-server; cold boot the emulator if needed
'Error connecting to the service protocol'	The app still installed; restart adb, or use flutter run --release

Problem	Fix
App says it can't reach the server	Start the backend (npm run dev) and check the address in section 7
Kotlin incremental build errors on Windows	Keep kotlin.incremental=false in gradle.properties
Strange build errors in OneDrive folders	Move the project out of OneDrive, then flutter clean and flutter pub get